

Lane-Topology-Guided Motion Forecasting via Feasible Motion Primitive Selection

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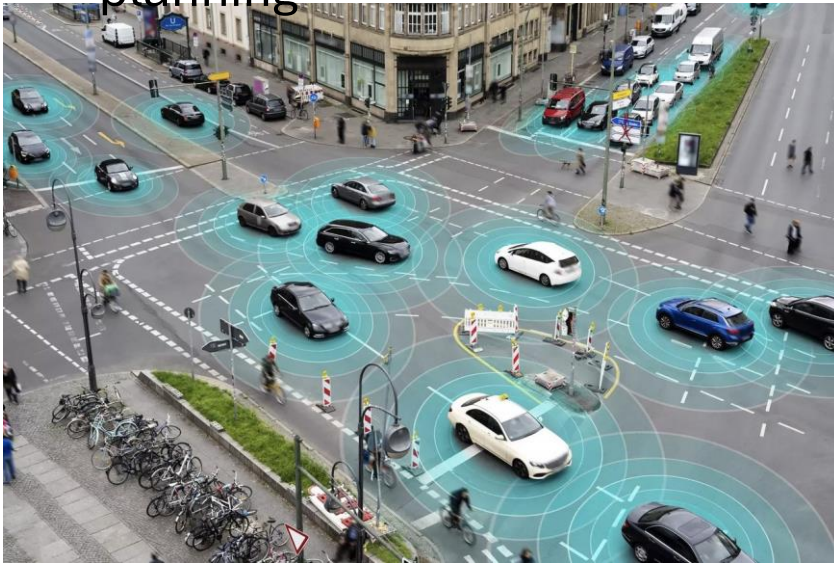
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Motion Forecasting : Key to Generalization in AV Systems

- Motion Forecasting (MF) enables generalization by anticipating future behaviors in unfamiliar environments
 - This predictive intent allows vehicles to proactively negotiate complex traffic safely
 - Examples: (1) Multi-agent nature in dense traffic, (2) Predictive context for safe proactive planning

(1)



(2)

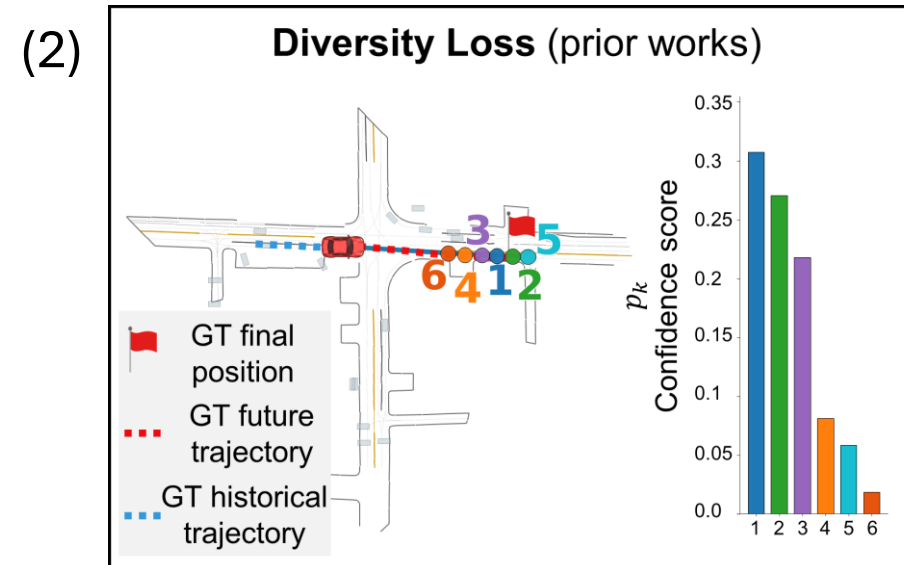
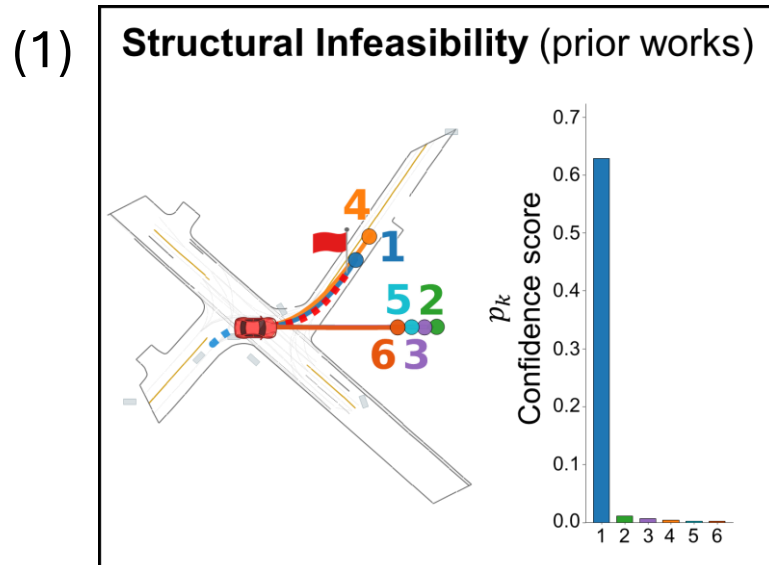


(1) <https://www.ed.tum.de/en/ed/news-single-view-start/article/the-hint-with-the-led-display-humans-and-automated-vehicles-in-road-traffic/>

(2) <https://www.youtube.com/watch?v=wdAoxZO92Zs> (Demonstrably Safe AI for Autonomous Driving, Waymo)

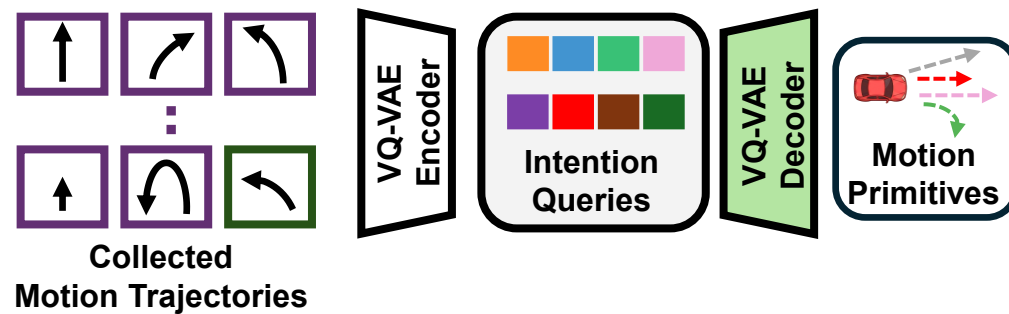
Challenge: Bridging Diversity and Structural Feasibility

- **Multimodal prediction** is essential to generate diverse plausible intentions capturing future stochasticity
- However, existing predictors often fail to focus on low-probability modes
 - (1) Prioritizing the primary ground-truth, **ignoring structural constraints** (e.g., lane topology)
 - (2) Training bias triggers mode collapse and **diversity loss**

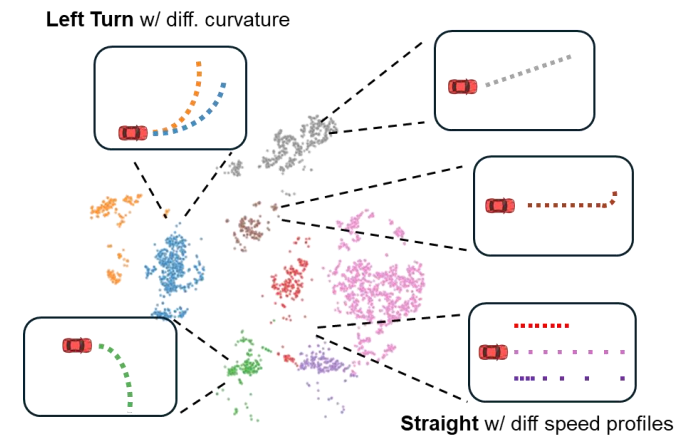


Design Principle 1 : Preserving Spatiotemporal Dynamics

- To secure diverse hypotheses, we **learn discrete intention queries** using NSVQ-based VQ-VAE
 - We preserve full spatiotemporal dynamics of driving behaviors beyond simple endpoints
 - This structural codebook captures interpretable driving patterns with varying curvatures and speeds



Architecture: VQ-VAE pipeline

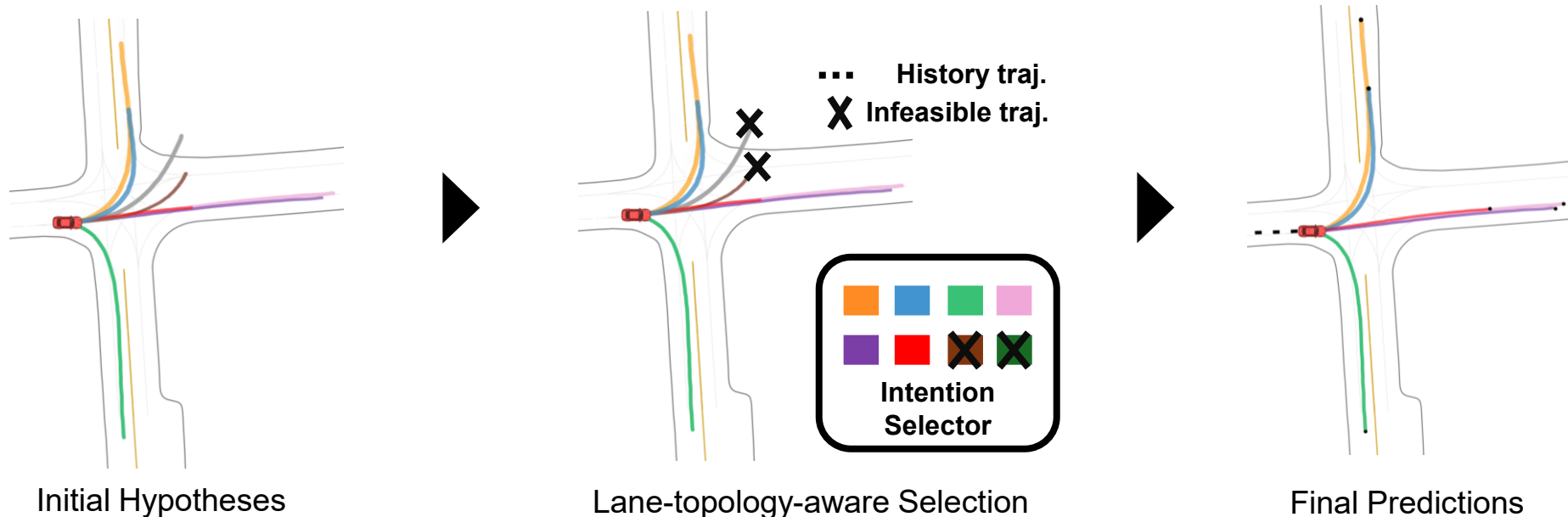


Learned primitives : Clustered intentions and motion primitives in latent space



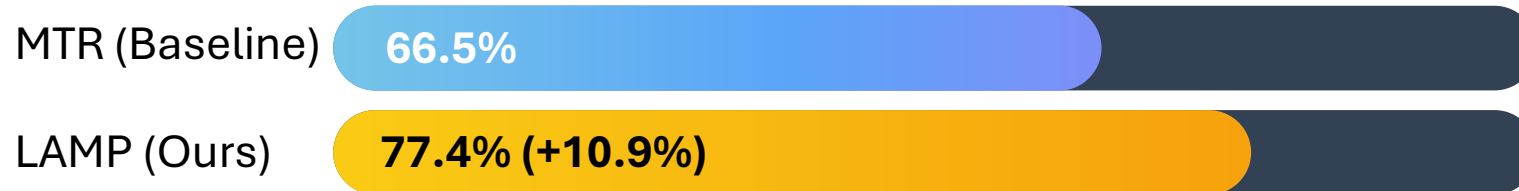
Design Principle 2 : Enforcing Structural Constraints

- To secure reliability, we **actively filter infeasible trajectories** before the final decoding stage
 - Specifically, we inject reachable-lane sets from HD maps and define a soft teacher distribution
 - The selector autonomously filters unreachable queries, prioritizing structural compliance
 - This internal guidance guarantees physically feasible predictions, directly enhancing downstream planning safety

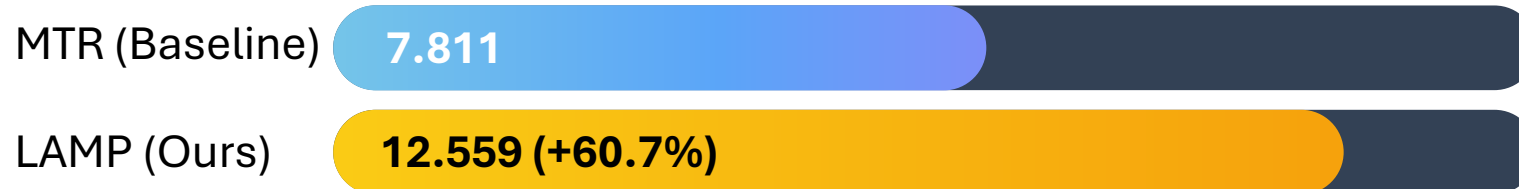


Quantitative Results Comparison

Feasibility Rate (FR) Comparison (%) ↑



Diversity-while-Feasible (DwF) ↑



Prediction Accuracy (b-minFDE₆) ↓



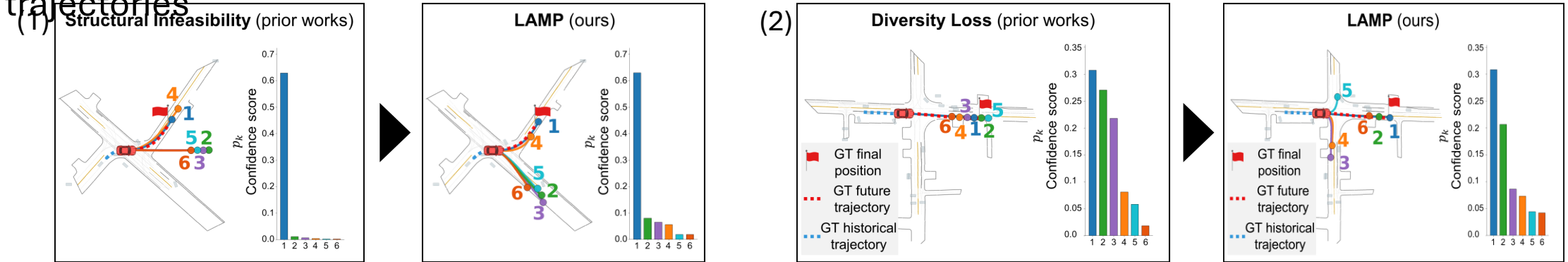
- LAMP achieves higher FR and DwF compared to the baseline, ensuring **varied and valid trajectory hypotheses**

- These gains are secured with only a marginal trade-off in prediction accuracy



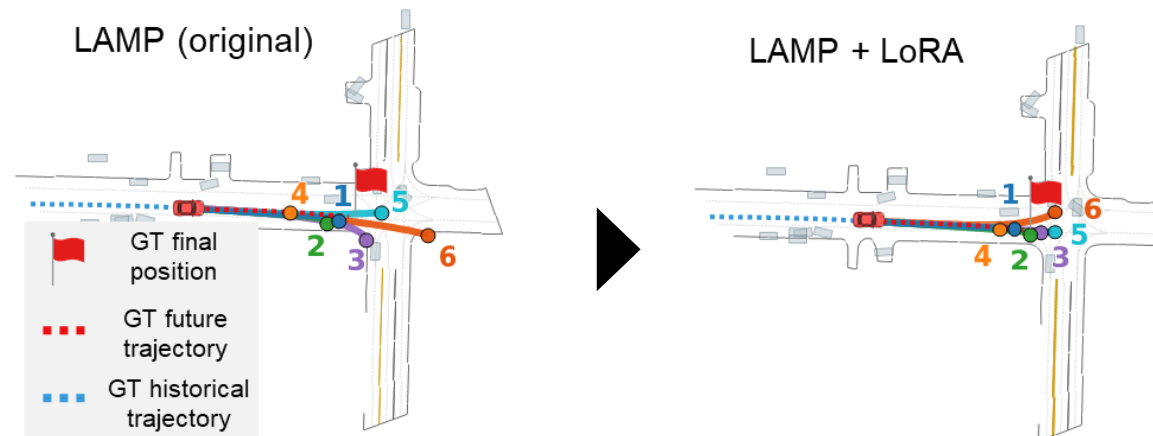
Qualitative Results Comparison

- Infeasible off-road predictions are replaced with (1) structurally compliant and (2) diverse trajectories



Findings: Map-adaptive motion primitives trade-off feasibility and diversity

- Applying LoRA for map-conditioned reconstruction improves lane compliance but induces mode collapse

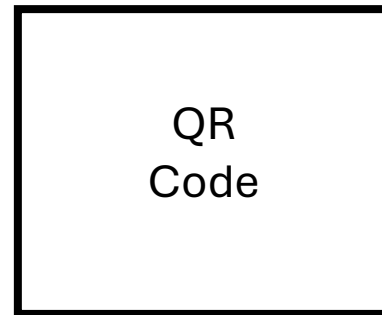


Conclusion

- **LAMP** enables diverse and lane-compliant forecasting by integrating discrete motion primitives with lane-topology constraints

Future Work

- We will extend LAMP to social interactions and evaluate downstream planning in closed-loop simulators



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